

Chapter 5

Normalization of Database Tables

Database Systems:
Design, Implementation, and
Management, Sixth Edition, Rob and
Coronel

Database Tables and Normalization

☐ Normalization

- Process for evaluating and correcting table structures to minimize data redundancies
 - ☐ helps eliminate data anomalies
- Works through a series of stages called normal forms:
 - ☐ Normal form (1NF)
 - ☐ Second normal form (2NF)
 - ☐ Third normal form (3NF)

Database Tables and Normalization

- 2NF is better than 1NF; 3NF is better than 2NF
- For most business database design purposes, 3NF is highest we need to go in the normalization process
- Highest level of normalization is not always most desirable

The Need for Normalization

- ❑ Example: company that manages building projects
 - Charges its clients by billing hours spent on each contract
 - Hourly billing rate is dependent on employee's position
 - Periodically, a report is generated that contains information displayed in Table 5.1

A Sample Report Layout

TABLE 5.1 A SAMPLE REPORT LAYOUT

PROJ. NUM.	PROJECT NAME	EMPLOYEE NUMBER	EMPLOYEE NAME	JOB CLASS.	CHG/HOUR	HOURS BILLED	TOTAL CHARGE
15	Evergreen	103	June E. Arbough	Elec. Engineer	\$84.50	23.8	\$2,011.10
		101	John G. News	Database Designer	\$105.00	19.4	\$2,037.00
		105	Alice K. Johnson *	Database Designer	\$105.00	35.7	\$3,748.50
		106	William Smithfield	Programmer	\$35.75	12.6	\$450.45
		102	David H. Senior	Systems Analyst	\$96.75	23.8	\$2,302.65
Subtotal							\$10,549.70
18	Amber Wave	114	Annelise Jones	Applications Designer	\$48.10	24.6	\$1,183.26
		118	James J. Frommer	General Support	\$18.36	45.3	\$831.71
		104	Anne K. Ramoras *	Systems Analyst	\$96.75	32.4	\$3,135.70
		112	Darlene M. Smithson	DSS Analyst	\$45.95	44.0	\$2,021.80
Subtotal							\$7,171.47
22	Rolling Tide	105	Alice K. Johnson	Database Designer	\$105.00	64.7	\$6,793.50
		104	Anne K. Ramoras	Systems Analyst	\$96.75	48.4	\$4,682.70
		113	Delbert K. Joenbrood *	Applications Designer	\$48.10	23.6	\$1,135.16
		111	Geoff B. Wabash	Clerical Support	\$26.87	22.0	\$591.14
		106	William Smithfield	Programmer	\$35.75	12.8	\$457.60
Subtotal							\$13,660.10
25	Starflight	107	Maria D. Alonzo	Programmer	\$35.75	24.6	\$879.45
		115	Travis B. Bawangi	Systems Analyst	\$96.75	45.8	\$4,431.15
		101	John G. News *	Database Designer	\$105.00	56.3	\$5,911.50
		114	Annelise Jones	Applications Designer	\$48.10	33.1	\$1,592.11
		108	Ralph B. Washington	Systems Analyst	\$96.75	23.6	\$2,283.30
		118	James J. Frommer	General Support	\$18.36	30.5	\$559.98
		112	Darlene M. Smithson	DSS Analyst	\$45.95	41.4	\$1,902.33
Subtotal							\$17,559.82
Total							\$48,941.09
* Indicates project leader							

A Table in the Report Format

FIGURE 5.1 A TABLE IN THE REPORT FORMAT

Table name: RPT_FORMAT				Database name: Ch05_ConstructCo			
	PROJ_NUM	PROJ_NAME	EMP_NUM	EMP_NAME	JOB_CLASS	CHG_HOUR	HOURS
► 15		Evergreen	103	June E. Arbough	Elect. Engineer	\$84.50	23.8
			101	John G. News	Database Designer	\$105.00	19.4
			105	Alice K. Johnson *	Database Designer	\$105.00	35.7
			106	William Smithfield	Programmer	\$35.75	12.6
			102	David H. Senior	Systems Analyst	\$96.75	23.8
18		Amber Wave	114	Annelise Jones	Applications Designer	\$48.10	24.6
			118	James J. Frommer	General Support	\$18.36	45.3
			104	Anne K. Ramoras *	Systems Analyst	\$96.75	32.4
			112	Darlene M. Smithson	DSS Analyst	\$45.95	44.0
22		Rolling Tide	105	Alice K. Johnson	Database Designer	\$105.00	64.7
			104	Anne K. Ramoras	Systems Analyst	\$96.75	48.4
			113	Delbert K. Joenbrood *	Applications Designer	\$48.10	23.6
			111	Geoff B. Wabash	Clerical Support	\$26.87	22.0
			106	William Smithfield	Programmer	\$35.75	12.8
25		Starflight	107	Maria D. Alonzo	Programmer	\$35.75	24.6
			115	Travis B. Bawangi	Systems Analyst	\$96.75	45.8
			101	John G. News *	Database Designer	\$105.00	56.3
			114	Annelise Jones	Applications Designer	\$48.10	33.1
			108	Ralph B. Washington	Systems Analyst	\$96.75	23.6
			118	James J. Frommer	General Support	\$18.36	30.5
			112	Darlene M. Smithson	DSS Analyst	\$45.95	41.4

The Need for Normalization

- ❑ Structure of data set in Figure 5.1 does not handle data very well
- ❑ The table structure *appears* to work; report is generated with ease
- ❑ Unfortunately, the report may yield different results, depending on what data anomaly has occurred

Conversion to First Normal Form

❑ Repeating group

- Derives its name from the fact that a group of multiple (related) entries can exist for any *single* key attribute occurrence
- ❑ Relational table must not contain repeating groups
- ❑ Normalizing the table structure will reduce these data redundancies
- ❑ Normalization is three-step procedure

Step 1: Eliminate the Repeating Groups

- ❑ Present data in a tabular format, where each cell has a single value and there are no repeating groups
- ❑ Eliminate repeating groups by eliminating nulls, making sure that each repeating group attribute contains an appropriate data value

Data Organization: First Normal Form

FIGURE 5.2 DATA ORGANIZATION: FIRST NORMAL FORM

Table name: DATA_ORG_1NF					Database name: Ch05_ConstructCo		
	PROJ_NUM	PROJ_NAME	EMP_NUM	EMP_NAME	JOB_CLASS	CHG_HOUR	HOURS
▶	15	Evergreen	103	June E. Arbough	Elect. Engineer	\$84.50	23.8
	15	Evergreen	101	John G. News	Database Designer	\$105.00	19.4
	15	Evergreen	105	Alice K. Johnson *	Database Designer	\$105.00	35.7
	15	Evergreen	106	William Smithfield	Programmer	\$35.75	12.6
	15	Evergreen	102	David H. Senior	Systems Analyst	\$96.75	23.8
	18	Amber Wave	114	Annelise Jones	Applications Designer	\$48.10	24.6
	18	Amber Wave	118	James J. Frommer	General Support	\$18.36	45.3
	18	Amber Wave	104	Anne K. Ramoras *	Systems Analyst	\$96.75	32.4
	18	Amber Wave	112	Darlene M. Smithson	DSS Analyst	\$45.95	44.0
	22	Rolling Tide	105	Alice K. Johnson	Database Designer	\$105.00	64.7
	22	Rolling Tide	104	Anne K. Ramoras	Systems Analyst	\$96.75	48.4
	22	Rolling Tide	113	Delbert K. Joenbrood *	Applications Designer	\$48.10	23.6
	22	Rolling Tide	111	Geoff B. Wabash	Clerical Support	\$26.87	22.0
	22	Rolling Tide	106	William Smithfield	Programmer	\$35.75	12.8
	25	Starflight	107	Maria D. Alonzo	Programmer	\$35.75	24.6
	25	Starflight	115	Travis B. Bawangi	Systems Analyst	\$96.75	45.8
	25	Starflight	101	John G. News *	Database Designer	\$105.00	56.3
	25	Starflight	114	Annelise Jones	Applications Designer	\$48.10	33.1
	25	Starflight	108	Ralph B. Washington	Systems Analyst	\$96.75	23.6
	25	Starflight	118	James J. Frommer	General Support	\$18.36	30.5
	25	Starflight	112	Darlene M. Smithson	DSS Analyst	\$45.95	41.4

Step 2: Identify the Primary Key

- ❑ Primary key must uniquely identify attribute value (PROJ_NUM is not unique)
- ❑ New key must be composed of PROJ_NUM and EMP_NUM

Step 3: Identify all Dependencies

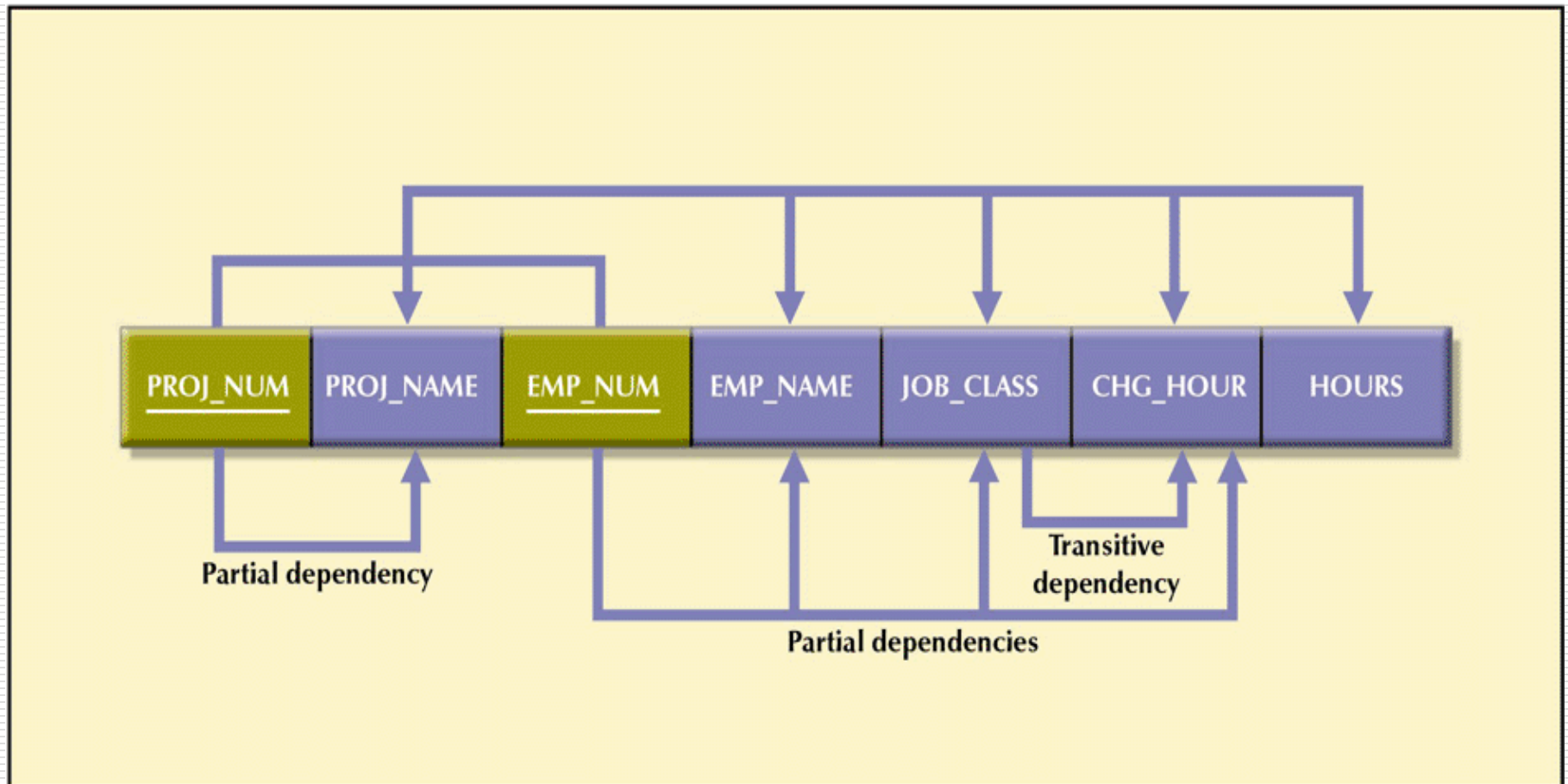
- ❑ Dependencies can be depicted with the help of a diagram
- ❑ Dependency diagram:
 - Depicts all dependencies found within a given table structure
 - Helpful in getting bird's-eye view of all relationships among a table's attributes
 - Use makes it much less likely that an important dependency will be overlooked

Step 3: Identify all Dependencies

- ❑ The arrows above the attributes indicate desirable dependencies i.e., ones that are based on the primary key
 - $\text{PROJ_NUM} + \text{EMP_NUM} \rightarrow \text{PROJ_NAME}, \text{EMP_NAME}, \text{JOB_CLASS}, \text{CHG_HOURS}, \text{HOURS}$
- ❑ The arrows below the attributes indicate less desirable dependencies
 - Partial dependencies – dependent on only part of the PK
 - ❑ $\text{PROJ_NUM} \rightarrow \text{PROJ_NAME}$
 - ❑ $\text{EMP_NUM} \rightarrow \text{EMP_NAME}, \text{JOB_CLASS}, \text{CHG_HOUR}$: only
 - Transitive dependencies – a dependency of one nonprime attribute on another nonprime attribute. They still yield data anomalies
 - ❑ $\text{JOB_CLASS} \rightarrow \text{CHG_HOUR}$

A Dependency Diagram: First Normal Form (1NF)

FIGURE 5.3 A DEPENDENCY DIAGRAM: FIRST NORMAL FORM (1NF)



First Normal Form

- ❑ Tabular format in which:
 - All key attributes are defined
 - There are no repeating groups in the table
 - All attributes are dependent on primary key
- ❑ All relational tables satisfy 1NF requirements
- ❑ Some tables contain partial dependencies
 - Dependencies based on only part of the primary key
 - Sometimes used for performance reasons, but should be used with caution
 - Still subject to data redundancies

Conversion to Second Normal Form

- ❑ Relational database design can be improved by converting the database into second normal form (2NF)
- ❑ Two step process

Step 1: Identify All Key Components

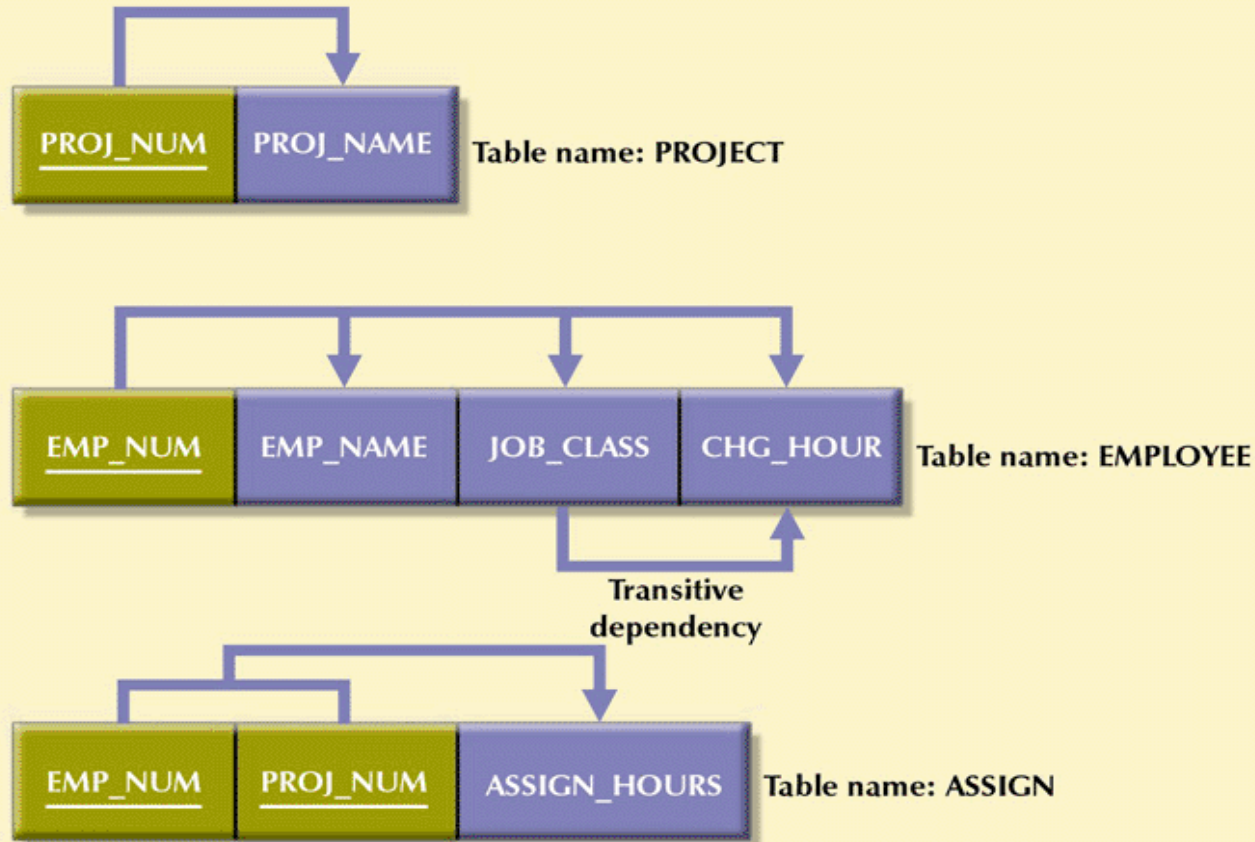
- ❑ Write each key component on separate line, and then write the original (composite) key on the last line
 - PROJ_NUM
 - EMP_NUM
 - PROJ_NUM EMP_NUM
- ❑ Each component will become the key in a new table

Step 2: Identify the Dependent Attributes

- ❑ Using the 1NF dependency diagram, determine which attributes are dependent on which other attributes
 - The dependencies are determined by examining the arrows below the diagram
 - PROJECT(**PROJ_NUM**, PROJ_NAME)
 - EMPLOYEE(**EMP_NUM**, EMP_NAME, JOB_CLASS, CHG_HOURS)
 - ASSIGN(**PROJ_NUM**, **EMP_NUM**, ASSIGN_HOURS)
- ❑ At this point, most anomalies have been eliminated

Second Normal Form (2NF) Conversion Results

FIGURE 5.4 SECOND NORMAL FORM (2NF) CONVERSION RESULTS



Second Normal Form

- ❑ Table is in second normal form (2NF) if:
 - It is in 1NF and
 - It includes no partial dependencies:
 - ❑ No attribute is dependent on only a portion of the primary key

Conversion to Third Normal Form

- ❑ Data anomalies created are easily eliminated by completing three steps

Step 1: Identify Each New Determinant

- For every transitive dependency, write its determinant as a PK for a new table

■ **Determinant**

- Any attribute whose value determines other values within a row

Step 2: Identify the Dependent Attributes

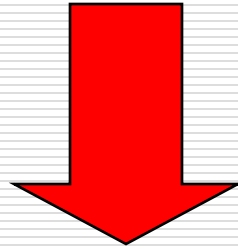
- ❑ Identify the attributes dependent on each determinant identified in Step 1 and identify the dependency
 - $\text{JOB_CLASS} \rightarrow \text{CHG_HOUR}$
- ❑ Name the table to reflect its contents and function
 - JOB

Step 3: Remove the Dependent Attributes from Transitive Dependencies

- ❑ Eliminate all dependent attributes in transitive relationship(s) from each table that has such a transitive relationship
- ❑ Draw a new dependency diagram to show all tables defined in Steps 1–3
- ❑ Check new tables and modified tables from Step 3 to make sure that each has a determinant and does not contain inappropriate dependencies

Step 3: Remove the Dependent Attributes from Transitive Dependencies

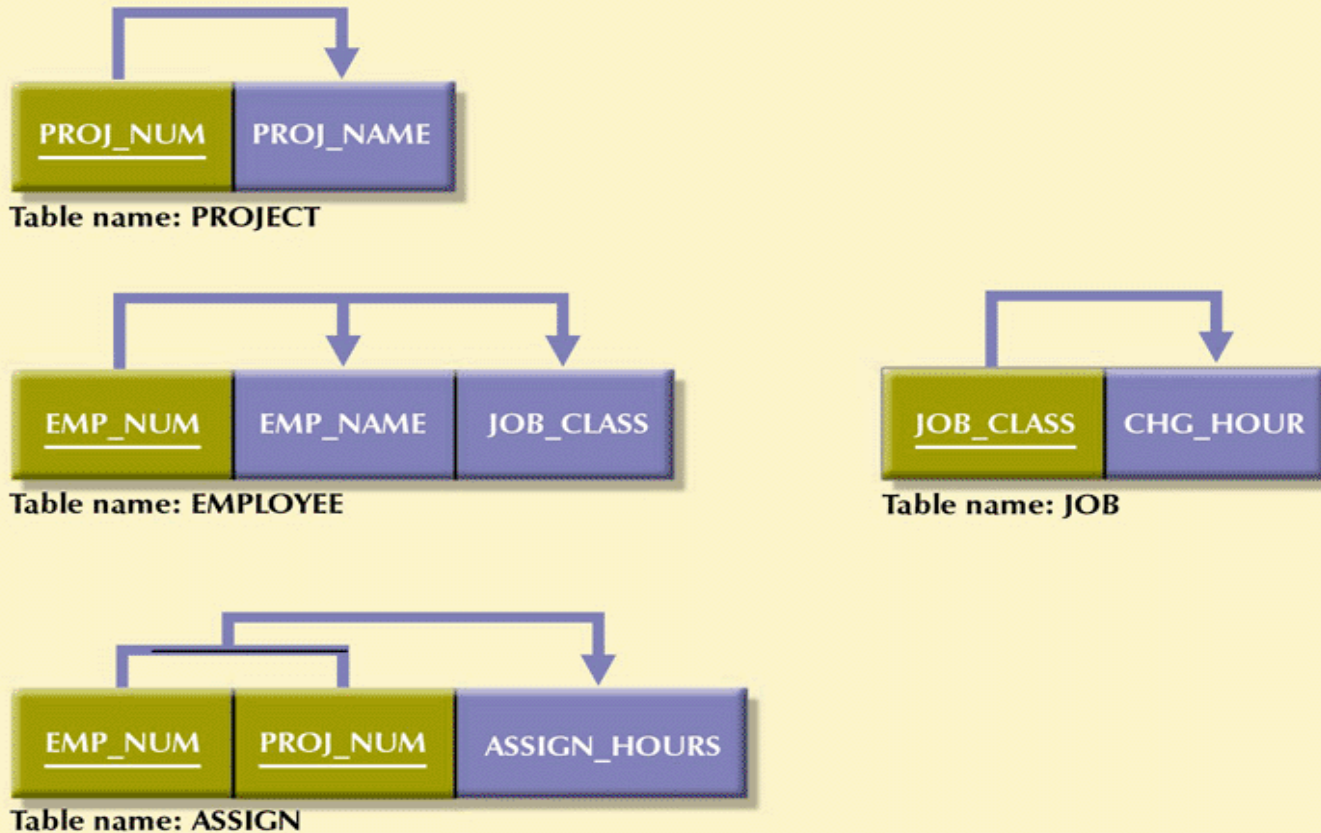
PROJECT(**PROJ_NUM**,PROJ_NAME)
EMPLOYEE(**EMP_NUM**,EMP_NAME,JOB_CLASS,CHG_HOURS)
ASSIGN(**PROJ_NUM,EMP_NUM**,ASSIGN_HOURS)



PROJECT(**PROJ_NUM**,PROJ_NAME)
ASSIGN(**PROJ_NUM,EMP_NUM**,ASSIGN_HOURS)
EMPLOYEE(**EMP_NUM**,EMP_NAME,JOB_CLASS)
JOB(**JOB_CLASS**,CHG_HOURS)

Third Normal Form (3NF) Conversion Results

FIGURE 5.5 THIRD NORMAL FORM (3NF) CONVERSION RESULTS



Third Normal Form

- ❑ A table is in third normal form (3NF) if:
 - It is in 2NF and
 - It contains no transitive dependencies

Improving the Design

- ❑ Table structures are cleaned up to eliminate the troublesome initial partial and transitive dependencies
- ❑ Normalization cannot, by itself, be relied on to make good designs
- ❑ It is valuable because its use helps eliminate data redundancies

Improving the Design

❑ PK assignment

- JOB_CLASS is entered into the EMPLOYEE table for each row. There is still potential for violation of referential integrity if one record has Database Designer and another DB Designer
- Thus, we add a JOB_CODE attribute
 - ❑ $\text{JOB_CODE} \rightarrow \text{JOB_CLASS}, \text{CHG_HOUR}$
 - ❑ This produces a transitive dependency of $\text{JOB_CLASS} \rightarrow \text{CHG_HOUR}$ if you assume that JOB_CODE is a PK
 - ❑ The benefit of reducing referential integrity errors outweighs the transitive dependency

Improving the Design

❑ Naming conventions

- CHG_HOUR changed to JOB_CHG_HOUR since it is part of the JOB table
- JOB_CLASS is replaced with JOB_DESCRIPTION as it gives a better indication of what the field contains (arguable)
- HOURS changed to ASSIGN_HOURS

❑ Attribute atomicity

- Replace EMP_NAME with fields for first and last name as well as initial

Improving the Design

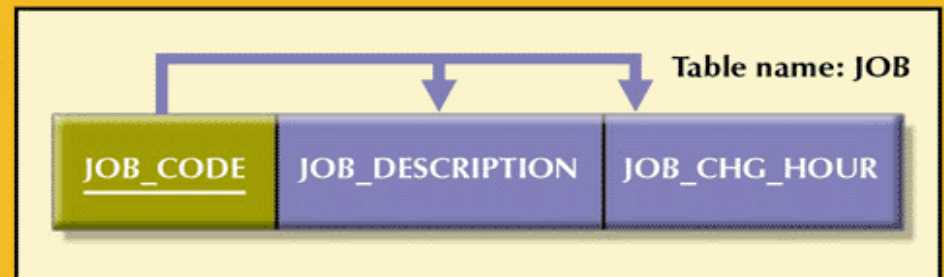
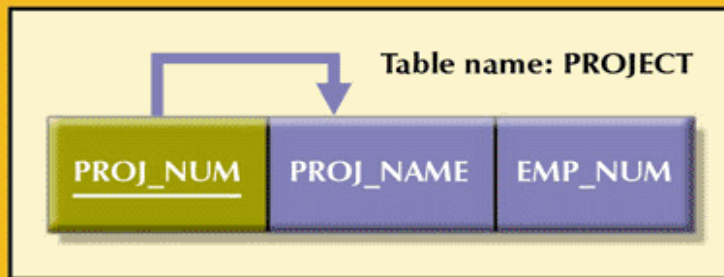
- ❑ Adding attributes
 - In the real world, the EMPLOYEE table would have many more attributes – YTD gross salary, social security and medicare payments, hire date, etc
- ❑ Adding relationships
 - By using EMP_NUM as a foreign key in PROJECT, we can easily associate all information about a project's manager with a project
- ❑ Refining PKs
 - It would be better to use a key such as an automatically generated sequential number called ASSIGN_NUM as a PK rather than EMP_NUM+PROJ_NUM for the ASSIGN table
 - ❑ If an employee makes two entries in the table for the same project, entity integrity is violated with the composite key
 - ❑ EMP_NUM and PROJ_NUM would still be used as FKs

Improving the Design

- ❑ Maintaining historical accuracy
 - Writing the job charge per hour into the ASSIGN table, as ASSIGN_CHG_HOUR, is crucial to maintain historical accuracy of the data
 - ❑ JOB_CHG_HOUR will change over time, we need to know the charge at the time the work was performed
- ❑ Using derived attributes
 - ❑ Storing derived attributes makes it easier to write the application software to generate the desired results and save time in generating the report
- ❑ We now have
PROJECT(**PROJ_NUM**, PROJ_NAME, EMP_NUM)
ASSIGN(**ASSIGN_NUM**, ASSIGN_DATE, PROJ_NUM, EMP_NUM,
ASSIGN_HOURS, ASSIGN_CHG_HOUR, ASSIGN_CHARGE)
EMPLOYEE(**EMP_NUM**, EMP_LNAME, EMP_FNAME, EMP_INITIAL,
EMP_HIREDATE, JOB_CLASS)
JOB(**JOB_DESCRIPTION**, JOB_CHG_HOUR)

The Completed Database

FIGURE 5.6 THE COMPLETED DATABASE



Database name: Ch05_ConstructCo

Table name: PROJECT

		PROJ_NUM	PROJ_NAME	EMP_NUM
▶	+	15	Evergreen	105
	+	18	Amber Wave	104
	+	22	Rolling Tide	113
	+	25	Starflight	101

Table name: JOB

		JOB_CODE	JOB_DESCRIPTION	JOB_CHG_HOUR
▶	+	500	Programmer	\$35.75
	+	501	Systems Analyst	\$96.75
	+	502	Database Designer	\$105.00
	+	503	Electrical Engineer	\$84.50
	+	504	Mechanical Engineer	\$67.90
	+	505	Civil Engineer	\$55.78
	+	506	Clerical Support	\$26.87
	+	507	DSS Analyst	\$45.95
	+	508	Applications Designer	\$48.10
	+	509	Bio Technician	\$34.55
	+	510	General Support	\$18.36

The Completed Database

FIGURE 5.6 THE COMPLETED DATABASE (CONTINUED)

Table name: ASSIGN

Table name: ASSIGN Database name: Ch05_ConstructCo

ASSIGN_NUM	ASSIGN_DATE	PROJ_NUM	EMP_NUM	ASSIGN_HOURS	ASSIGN_CHG_HOUR	ASSIGN_CHARGE
1001	04-Mar-04	15	103	2.6	\$84.50	\$219.70
1002	04-Mar-04	18	118	1.4	\$18.36	\$25.70
1003	05-Mar-04	15	101	3.6	\$105.00	\$378.00
1004	05-Mar-04	22	113	2.5	\$48.10	\$120.25
1005	05-Mar-04	15	103	1.9	\$84.50	\$160.55
1006	05-Mar-04	25	115	4.2	\$96.75	\$406.35
1007	05-Mar-04	22	105	5.2	\$105.00	\$546.00
1008	05-Mar-04	25	101	1.7	\$105.00	\$178.50
1009	05-Mar-04	15	105	2.0	\$105.00	\$210.00
1010	06-Mar-04	15	102	3.8	\$96.75	\$367.65
1011	06-Mar-04	22	104	2.6	\$96.75	\$251.55
1012	06-Mar-04	15	101	2.3	\$105.00	\$241.50
1013	06-Mar-04	25	114	1.8	\$48.10	\$86.58
1014	06-Mar-04	22	111	4.0	\$26.87	\$107.48
1015	06-Mar-04	25	114	3.4	\$48.10	\$163.54
1016	06-Mar-04	18	112	1.2	\$45.95	\$55.14
1017	06-Mar-04	18	118	2.0	\$18.36	\$36.72
1018	06-Mar-04	18	104	2.6	\$96.75	\$251.55
1019	06-Mar-04	15	103	3.0	\$84.50	\$253.50
1020	07-Mar-04	22	105	2.7	\$105.00	\$283.50
1021	08-Mar-04	25	108	4.2	\$96.75	\$406.35
1022	07-Mar-04	25	114	5.8	\$48.10	\$278.98
1023	07-Mar-04	22	106	2.4	\$35.75	\$85.80

Table name: EMPLOYEE

Table name: EMPLOYEE

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_HIREDATE	JOB_CODE
101	News	John	G	08-Nov-98	502
102	Senior	David	H	12-Jul-87	501
103	Arbough	June	E	01-Dec-94	503
104	Ramoras	Anne	K	15-Nov-85	501
105	Johnson	Alice	K	01-Feb-91	502
106	Smithfield	William		22-Jun-03	500
107	Alonzo	Maria	D	10-Oct-91	500
108	Washington	Ralph	B	22-Aug-89	501
109	Smith	Larry	WV	18-Jul-95	501
110	Olenko	Gerald	A	11-Dec-93	505
111	Wabash	Geoff	B	04-Apr-89	506
112	Smithson	Darlene	M	23-Oct-92	507
113	Joerbrood	Delbert	K	15-Nov-94	508
114	Jones	Annelise		20-Aug-91	508
115	Bawangi	Travis	B	25-Jan-90	501
116	Pratt	Gerald	L	05-Mar-95	510
117	Williamson	Angie	H	19-Jun-94	509
118	Frommer	James	J	04-Jan-04	510

Limitations on System-Assigned Keys

- ❑ System-assigned primary key may not prevent confusing entries
- ❑ Data entries in Table 5.2 are inappropriate because they duplicate existing records
 - Yet there has been no violation of either entity integrity or referential integrity
 - Ensure unique job descriptions by making a unique index on that field
 - Trade-off between design integrity and flexibility- manager may want an employee to make multiple entries per day

TABLE 5.2 DUPLICATE ENTRIES IN THE JOB TABLE

JOB_CODE	JOB_DESCRIPTION	JOB_CHG_HOUR
511	Programmer	\$35.75
512	Programmer	\$35.75

The Boyce-Codd Normal Form (BCNF)

- ❑ Every determinant in the table is a candidate key
 - Has same characteristics as primary key, but for some reason, not chosen to be primary key
- ❑ If a table contains only one candidate key, the 3NF and the BCNF are equivalent
- ❑ BCNF can be violated only if the table contains more than one candidate key

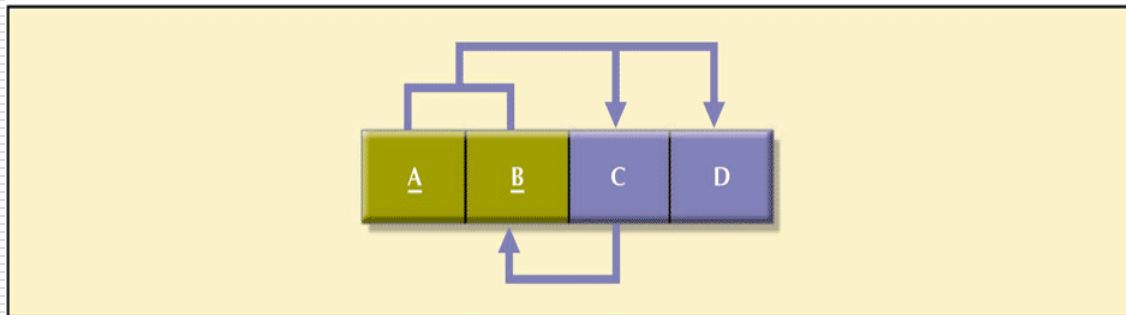
The Boyce-Codd Normal Form

- ❑ A table is in BCNF if every determinant in the table is a candidate key
 - BCNF is violated if a table has more than one candidate key
- ❑ Most designers consider the Boyce-Codd normal form (BCNF) as a special case of 3NF
 - A table is in 3NF if it is in 2NF and there are no transitive dependencies
 - ❑ A transitive dependency exists when one nonprime attribute is dependent on another nonprime attribute
 - A table can be in 3NF and not be in BCNF if a nonkey attribute is the determinant of a key attribute

A Table That is in 3NF but not in BCNF

- ❑ Note these functional dependencies
 - $A+B \rightarrow C, D$
 - $C \rightarrow B$ (nonkey determines part of the key)
- ❑ The table has no partial or transitive dependencies so it is in 3NF

FIGURE 5.7 A TABLE THAT IS IN 3NF BUT NOT IN BCNF

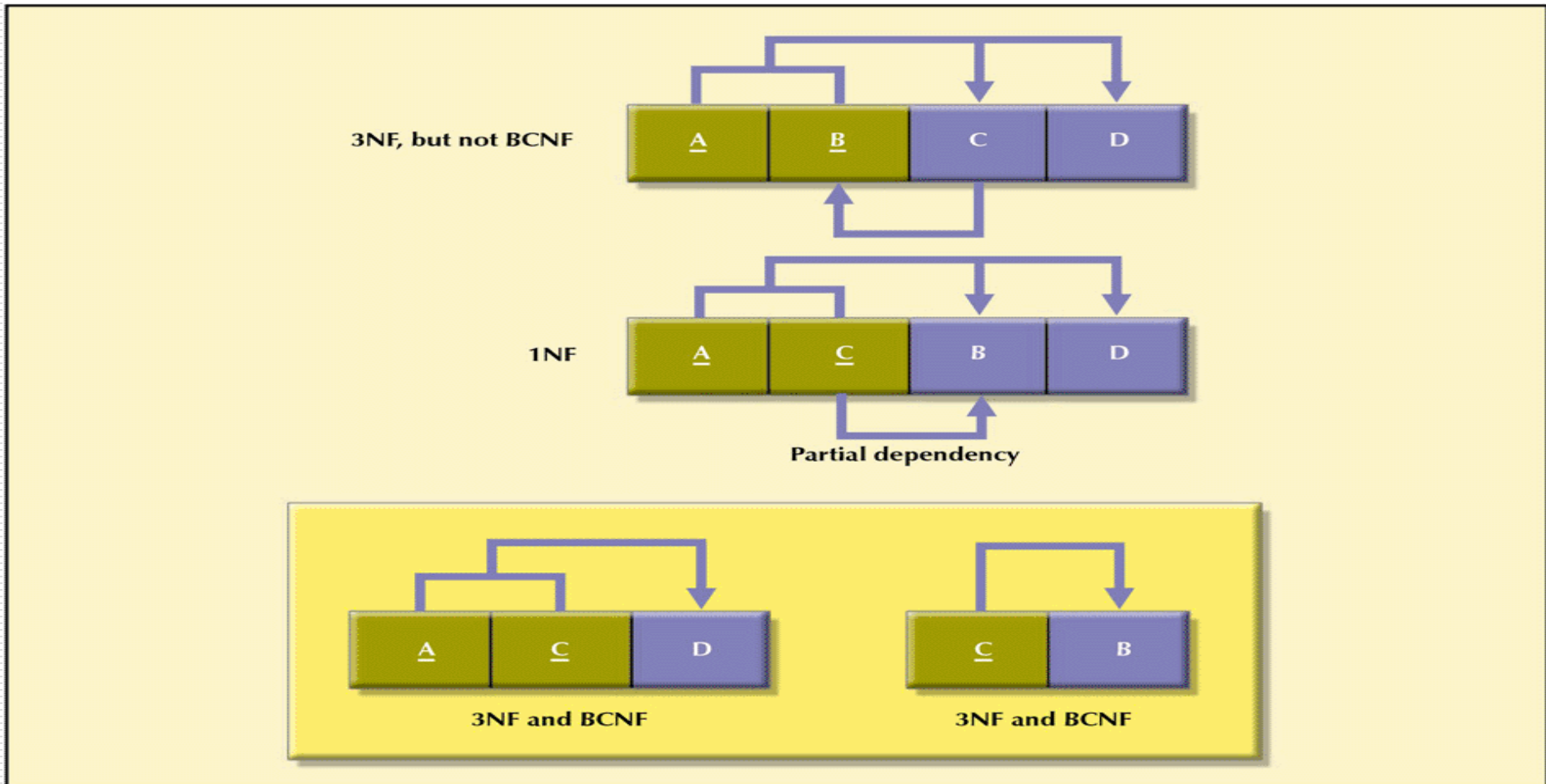


Decomposition to BCNF

- ❑ Change the PK to A+C (since $C \rightarrow B$)
 - the table is in 1NF since there is a partial dependency $C \rightarrow B$
 - Decompose table as before

Decomposition to BCNF

FIGURE 5.8 DECOMPOSITION TO BCNF



Sample Data for a BCNF Conversion

TABLE 5.3 SAMPLE DATA FOR A BCNF CONVERSION

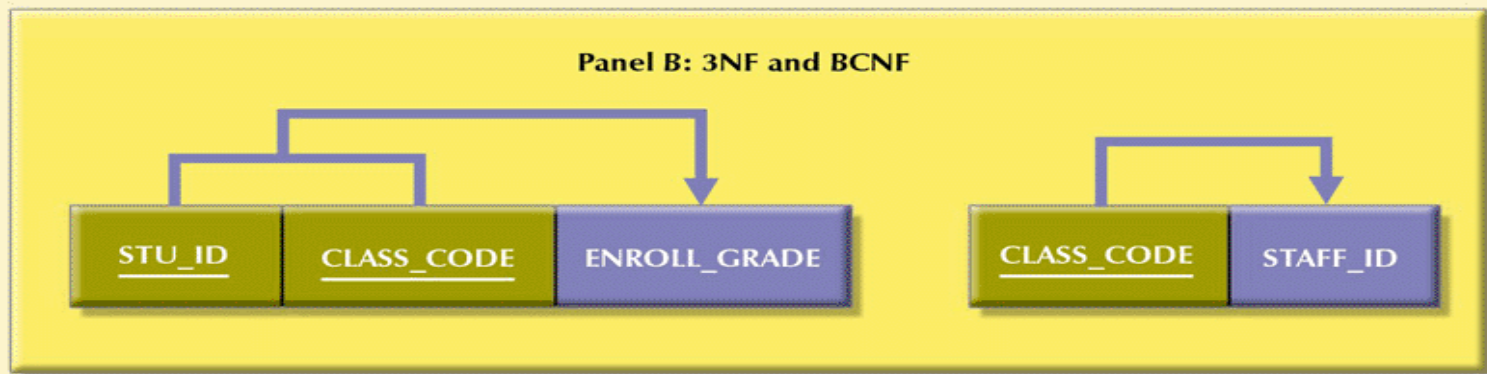
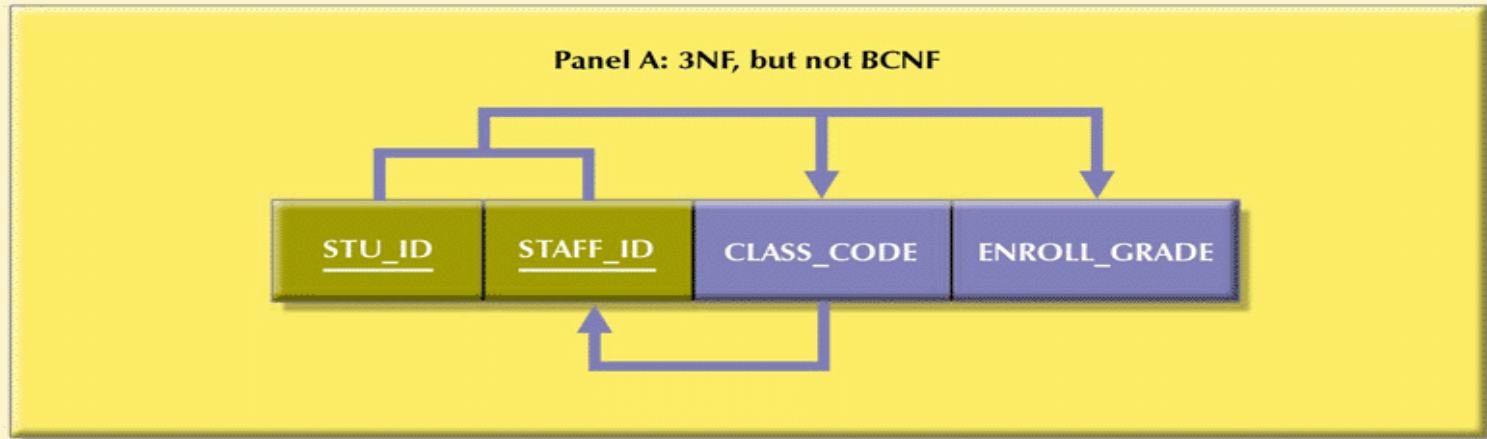
STU_ID	STAFF_ID	CLASS_CODE	ENROLL_GRADE
125	25	21334	A
125	20	32456	C
135	20	28458	B
144	25	27563	C
144	20	32456	B

Sample Data for a BCNF Conversion

- ❑ CLASS_CODE identifies a class uniquely (might represent course and section)
- ❑ A student can take many classes
- ❑ A staff member can teach many classes but each class is taught by only one staff member
- ❑ In Panel A (next slide) an anomaly can occur when
 - The staff member who teaches a course is changed. Each CLASS_CODE has to have the associated STAFF_ID updated
 - If a student drops a course, we can lose information about who taught the course

Another BCNF Decomposition

FIGURE 5.9 ANOTHER BCNF DECOMPOSITION



Normalization and Database Design

- ❑ Normalization should be part of design process
- ❑ Make sure that proposed entities meet required normal form before table structures are created
- ❑ Many real-world databases have been improperly designed or burdened with anomalies if improperly modified during course of time
- ❑ You may be asked to redesign and modify existing databases

Normalization and Database Design

☐ ER diagram

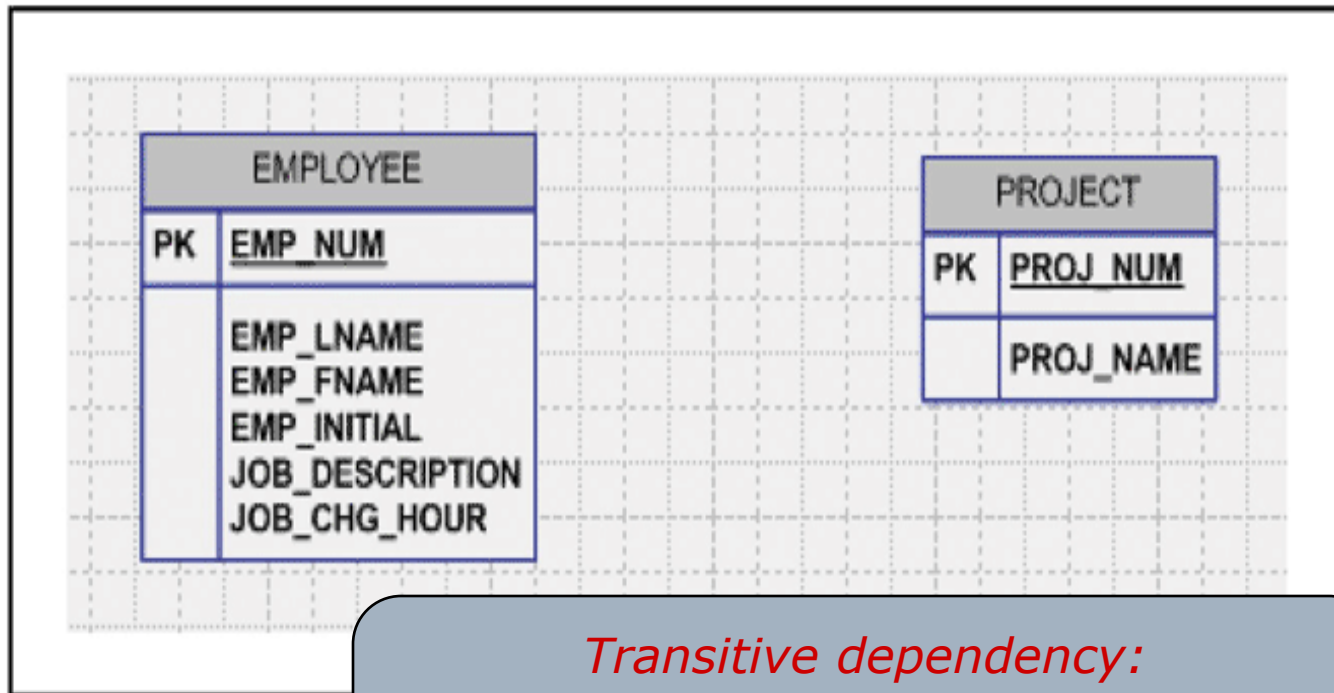
- Provides the big picture, or macro view, of an organization's data requirements and operations
- Created through an iterative process
 - ☐ Identifying relevant entities, their attributes and their relationship
 - ☐ Use results to identify additional entities and attributes

Normalization and Database Design

- ❑ Normalization procedures
 - Focus on the characteristics of specific entities
 - A micro view of the entities within the ER diagram
- ❑ Difficult to separate normalization process from ER modeling process
- ❑ Two techniques should be used concurrently

The Initial ERD for a Contracting Company

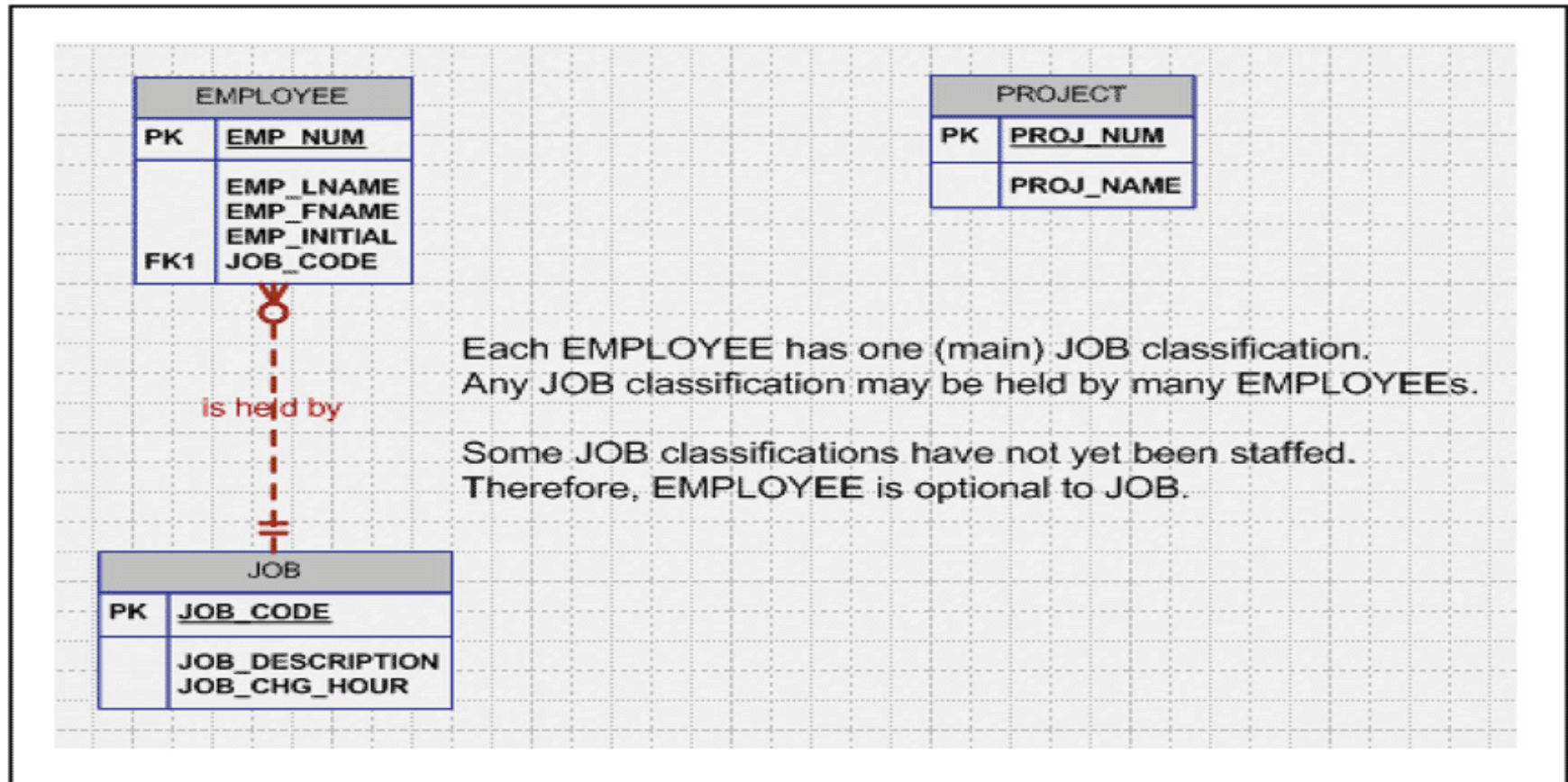
FIGURE 5.10 THE INITIAL ERD FOR A CONTRACTING COMPANY



Transitive dependency:
JOB_DESCRIPTION defines job classifications which in turn determine billing rates (JOB_CHG_HOUR)

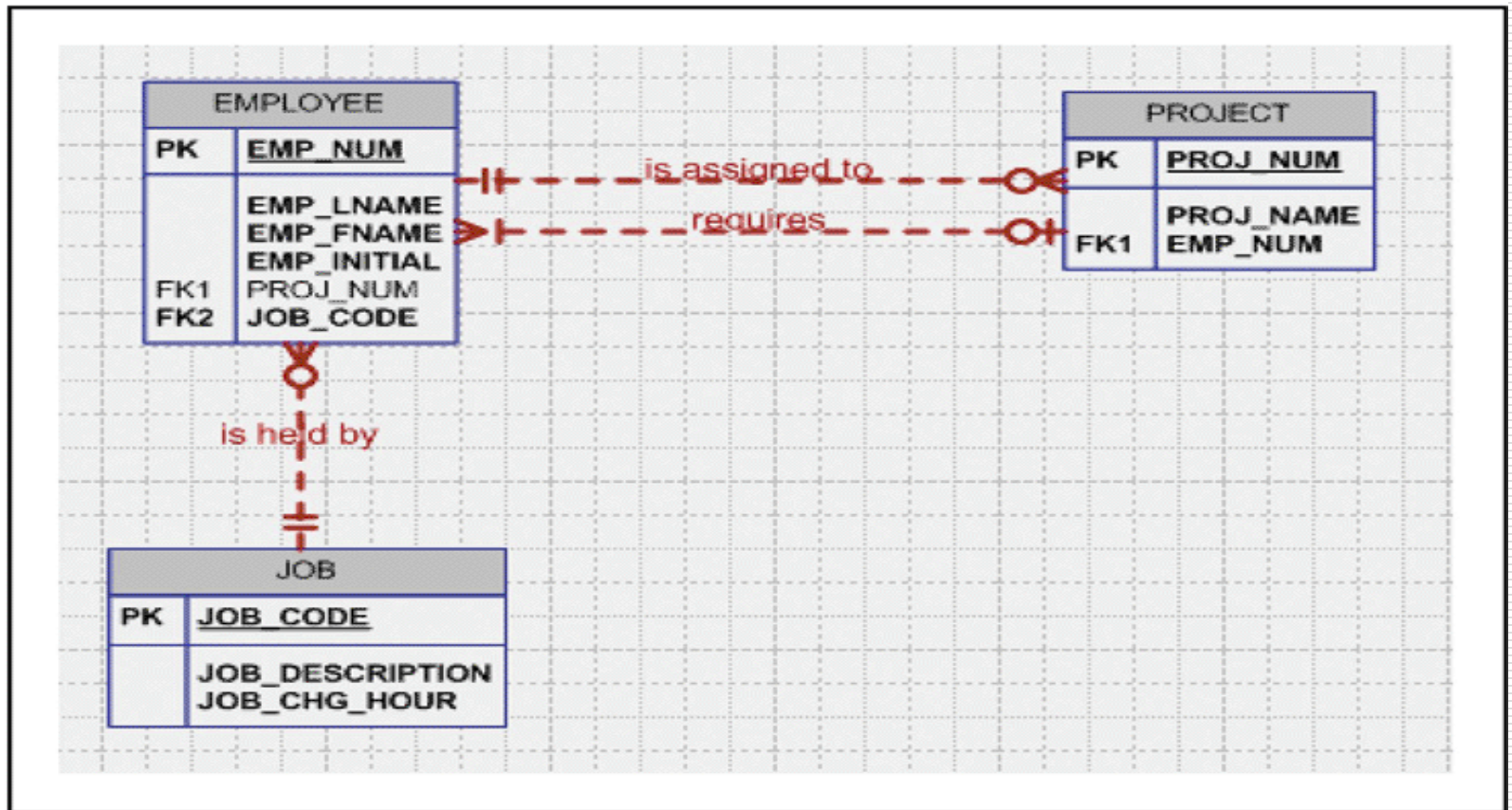
The Modified ERD for a Contracting Company

FIGURE 5.11 THE MODIFIED ERD FOR A CONTRACTING COMPANY



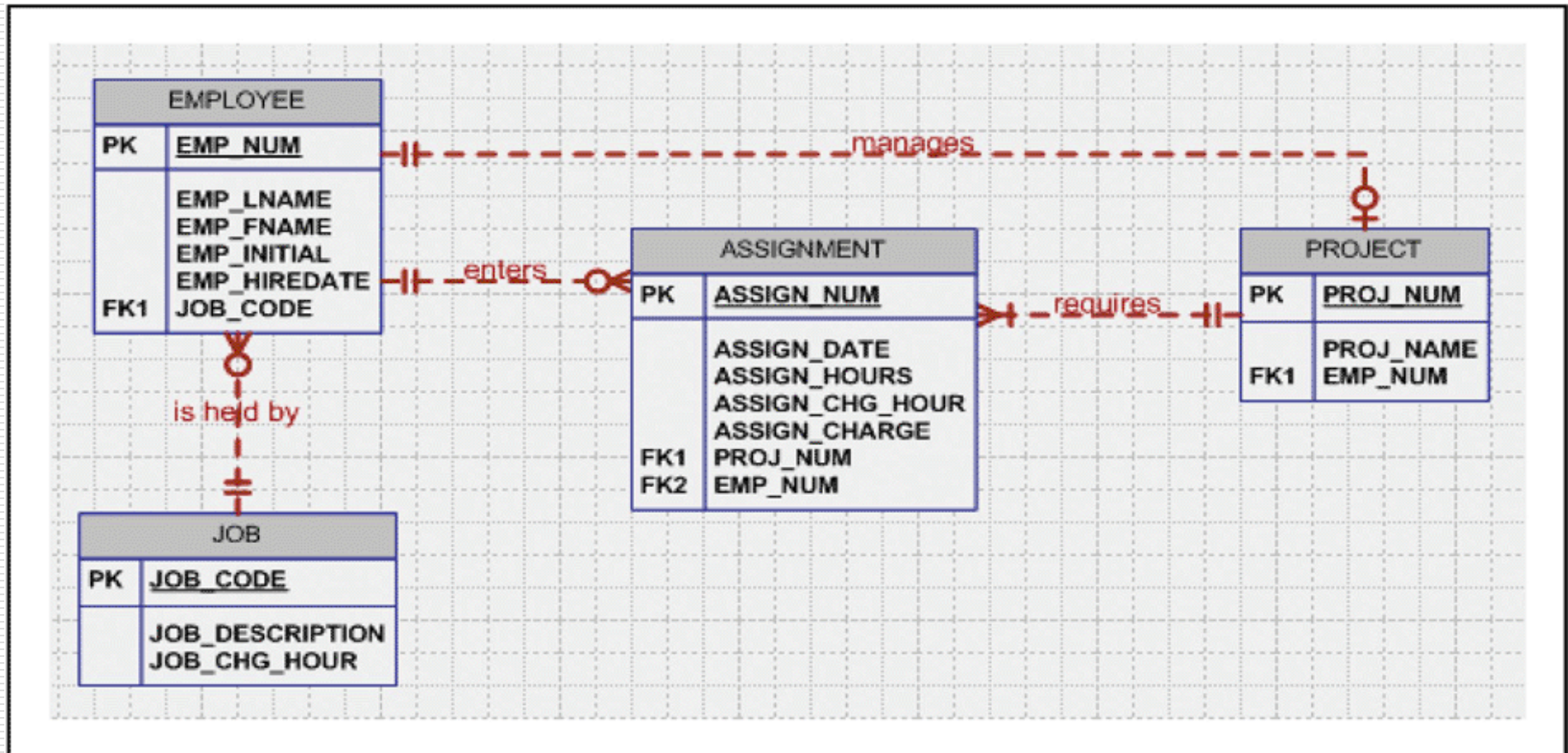
The Incorrect Representation of a M:N Relationship

FIGURE 5.12 THE INCORRECT REPRESENTATION OF A M:N RELATIONSHIP



The Final (Implementable) ERD for a Contracting Company

FIGURE 5.13 THE FINAL (IMPLEMENTABLE) ERD FOR A CONTRACTING COMPANY



The Implemented Database for the Contracting Company

FIGURE 5.14 THE IMPLEMENTED DATABASE FOR THE CONTRACTING COMPANY

Table name: EMPLOYEE

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_HIREDATE	JOB_CODE
101	News	John	G	08-Nov-98	S02
102	Senior	David	H	12-Jul-87	S01
103	Arbough	June	E	01-Dec-94	S03
104	Ramoras	Anne	K	15-Nov-85	S01
105	Johnson	Alice	K	01-Feb-91	S02
106	Smithfield	William		22-Jun-03	S00
107	Alonzo	Maria	D	10-Oct-91	S00
108	Washington	Ralph	B	22-Aug-89	S01
109	Smith	Larry	W	18-Jul-95	S01
110	Olenko	Gerald	A	11-Dec-93	S05
111	Wabash	Geoff	B	04-Apr-89	S06
112	Smithson	Darlene	M	23-Oct-92	S07
113	Joebrood	Delbert	K	15-Nov-94	S08
114	Jones	Annelise		20-Aug-91	S08
115	Bawangi	Travis	B	25-Jan-90	S01
116	Pratt	Gerald	L	05-Mar-95	S10
117	Williamson	Angie	H	19-Jun-94	S09
118	Frommer	James	J	04-Jan-04	S10

Database name: Ch05_ConstructCo

Table name: JOB

JOB_CODE	JOB_DESCRIPTION	JOB_CHG_HOUR
500	Programmer	\$35.75
501	Systems Analyst	\$96.75
502	Database Designer	\$105.00
503	Electrical Engineer	\$84.50
504	Mechanical Engineer	\$67.90
505	Civil Engineer	\$55.78
506	Clerical Support	\$26.87
507	DSS Analyst	\$45.95
508	Applications Designer	\$48.10
509	Bio Technician	\$34.55
510	General Support	\$18.36

Table name: PROJECT

PROJ_NUM	PROJ_NAME	EMP_NUM
19	Evergreen	105
18	Amber Wave	104
22	Rolling Tide	113
25	Starflight	101

Table name: ASSIGNMENT

ASSIGN_NUM	ASSIGN_DATE	PROJ_NUM	EMP_NUM	ASSIGN_HOURS	ASSIGN_CHG_HOUR	ASSIGN_CHARGE
1001	04-Mar-04	15	103	2.6	\$84.50	\$219.70
1002	04-Mar-04	18	118	1.4	\$18.36	\$25.70
1003	05-Mar-04	15	101	3.6	\$105.00	\$378.00
1004	05-Mar-04	22	113	2.5	\$48.10	\$120.25
1005	05-Mar-04	15	103	1.9	\$84.50	\$160.55
1006	05-Mar-04	25	115	4.2	\$96.75	\$406.35
1007	05-Mar-04	22	105	5.2	\$105.00	\$546.00
1008	05-Mar-04	25	101	1.7	\$105.00	\$178.50
1009	05-Mar-04	15	105	2.0	\$105.00	\$210.00
1010	06-Mar-04	15	102	3.8	\$96.75	\$367.65
1011	06-Mar-04	22	104	2.6	\$96.75	\$251.55
1012	06-Mar-04	15	101	2.3	\$105.00	\$241.50
1013	06-Mar-04	25	114	1.8	\$48.10	\$86.58
1014	06-Mar-04	22	111	4.0	\$26.87	\$107.48
1015	06-Mar-04	25	114	3.4	\$48.10	\$163.54
1016	06-Mar-04	18	112	1.2	\$45.95	\$55.14
1017	06-Mar-04	18	118	2.0	\$18.36	\$36.72
1018	06-Mar-04	18	104	2.6	\$96.75	\$251.55
1019	06-Mar-04	15	103	3.0	\$84.50	\$253.50
1020	07-Mar-04	22	105	2.7	\$105.00	\$283.50
1021	08-Mar-04	25	108	4.2	\$96.75	\$406.35
1022	07-Mar-04	25	114	5.8	\$48.10	\$278.98
1023	07-Mar-04	22	106	2.4	\$35.75	\$85.80

Higher-Level Normal Forms

- ❑ In some databases, multiple multivalued attributes exist
 - An employee can have multiple assignments and can also be involved in multiple service organizations(Red Cross, United Way)
 - Tables on next slide contain two sets of independent multivalued dependencies
 - ❑ Versions 1 and 2 can have null values so there isn't a candidate key
 - ❑ Version 3 is in 3NF but contains redundancies

Tables with Multivalued Dependencies

FIGURE 5.15 TABLES WITH MULTIVALUED DEPENDENCIES

Database name: Ch05_Service

Table name: VOLUNTEER_V1

	EMP_NUM	ORG_CODE	ASSIGN_NUM
▶	10123	RC	1
	10123	UW	3
	10123		4

Table name: VOLUNTEER_V3

	EMP_NUM	ORG_CODE	ASSIGN_NUM
▶	10123	RC	1
	10123	RC	3
	10123	UW	4

Table name: VOLUNTEER_V2

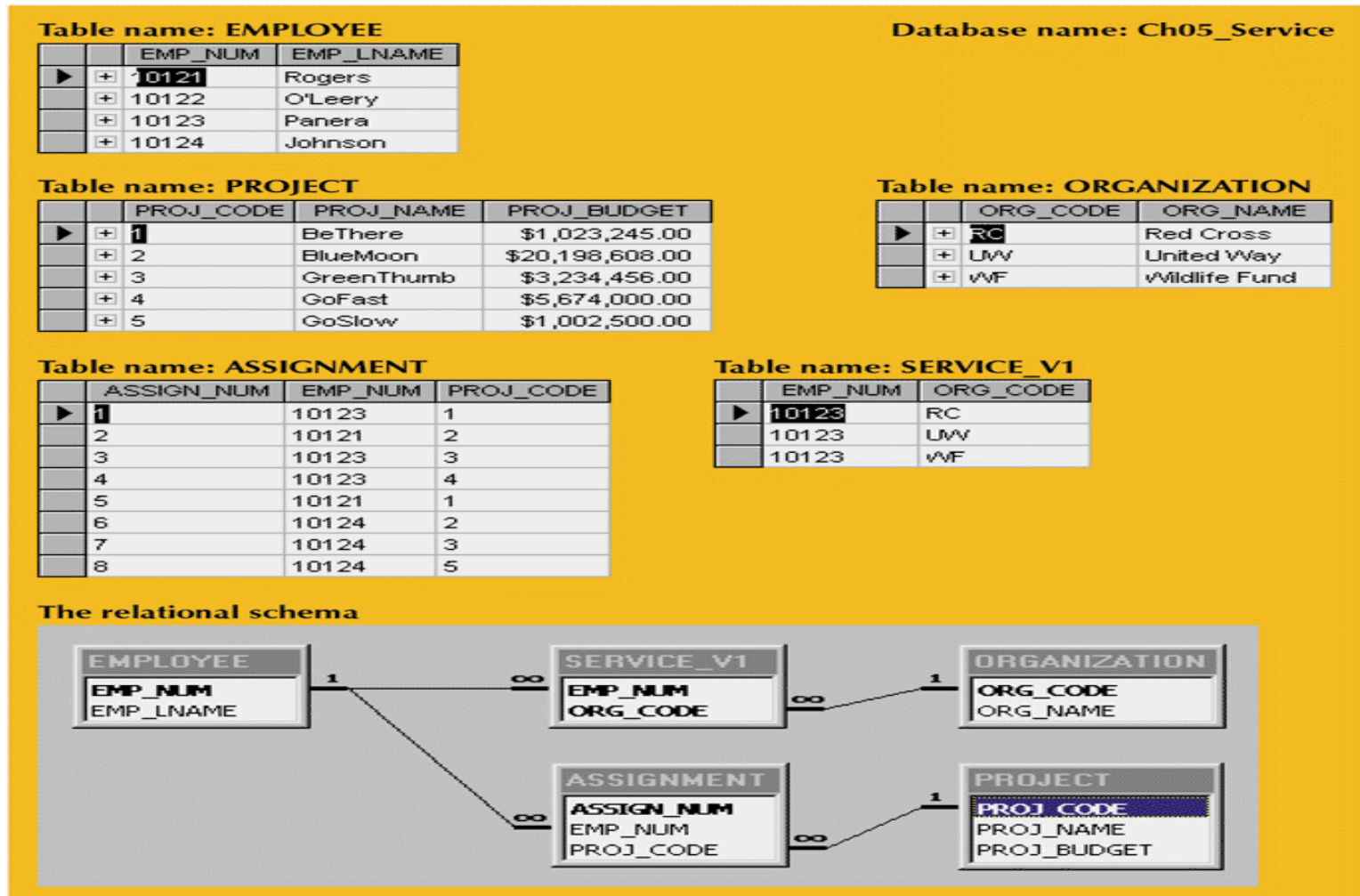
	EMP_NUM	ORG_CODE	ASSIGN_NUM
▶	10123	RC	
	10123	UW	
	10123		1
	10123		3
	10223		4

Fourth Normal Form

- ❑ Table is in fourth normal form (4NF) if
 - It is in 3NF
 - Has no multiple sets of multivalued dependencies
- ❑ 4NF is largely academic if tables conform to the following two rules:
 - All attributes are dependent on primary key but independent of each other
 - No row contains two or more multivalued facts about an entity

A Set of Tables in 4NF

FIGURE 5.16 A SET OF TABLES IN 4NF



Denormalization

- ❑ Creation of normalized relations is important database design goal
- ❑ Processing requirements should also be a goal
- ❑ If tables decomposed to conform to normalization requirements, then the number of database tables expands

Denormalization

- ❑ Joining larger number of tables takes additional disk input/output (I/O) operations and processing logic
 - Reduces system speed
- ❑ Conflicts among design efficiency, information requirements, and processing speed are often resolved through compromises that may include denormalization

Denormalization (continued)

- ❑ Unnormalized tables in a production database tend to have these defects:
 - Data updates are less efficient because programs that read and update tables must deal with larger tables
 - Indexing is much more cumbersome
 - Unnormalized tables yield no simple strategies for creating virtual tables known as *views*

Denormalization (continued)

- ❑ Use denormalization cautiously
- ❑ Understand why—under some circumstances—unnormalized tables are a better choice